

Plant Disease Detection and Pesticide Recommendation Using Deep Learning

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Abstract—Plant illnesses represent a considerable risk to the productivity of agriculture and the security of our food supply. This paper introduces an automated system utilizing deep learning for the identification of plant diseases and recommending pesticides. The system employs a modified EfficientNetB5 model for disease detection, trained on a dataset comprising labeled images featuring both healthy and diseased plants. Image preprocessing techniques were applied to enhance image quality and improve model performance. The system achieved an accuracy of over 96% in identifying common plant diseases. Furthermore, based on the identified disease, the system recommends suitable pesticides for treatment. This automated approach offers a fast and accurate method for early disease detection and intervention, ultimately contributing to improved crop management practices and increased agricultural yield.

Keywords—Agriculture productivity, plant disease identification, Plant Pesticide, Deep Learning (DL), EfficientNet-B5.

I. INTRODUCTION

Human life is governed by living and nonliving things among which species of plants are important for their vital functions. One of the essential resources in this world is plants and their existence is everywhere. While considering the protection of the environment, plants play a major role. Plant disease severity can be used to forecast the yield since it is a powerful limitation on evaluating disease stages. Recognizing plant disease using camera devices in mobile phone images proves a significant task. In this regard, we look for a reliable, fast, automatic, less costly, and accurate method of plant disease detection to overcome this situation. When farmers discover a plant disease, they typically use only chemical fertilizers to prevent further disease growth. This may have harmful effects on the plant and the person contacting the plant. Sometimes, it helps solve a problem with essential products, like plucking and burning a diseased leaf or organic fertilizers. All this depends on how much the leaves of a crop have been affected by a disease. Taking appropriate management steps benefits from early disease prediction. This marks a crucial advancement in managing disease proliferation and generating top-notch products to fulfill the global demand for food. This stage will help avoid further disease development in plants, and the outcome from plants will benefit the growing population. Identifying plant leaf diseases is most significant in increasing crop yield and growth. Manual methods for

plant disease identification are subjective, labour-intensive, and not scalable. They rely on visual inspection, leading to inaccuracies and delays in diagnosis, especially in regions with limited expertise. Deep learning methods offer a more objective and efficient alternative for plant disease identification.

A. Motivation of Research

Extinction in plants has become a significant problem nowadays as several plant species face it. The rate of extinction among plant species is reaching an alarming count. It is the responsibility of a botanist to find the plant that is nearing extinction prior to the change in climate and immediate solutions need to be developed. The images of leaves can be utilized for the examining work where automation of these methods is necessary for detection and classification processes. Identifying the diseases in a manual way requires more knowledge regarding plants and their varieties.

B. Significance of the Study

Agriculture falls because of crops being affected in many regions due to lack of maintenance which impacts the necessary needs. There are many threats to food that lacks a healthy infrastructure. In this term, the combination of agricultural disease and its spread is higher nowadays. Crop production is affected due to these diseases as pests infect them and products produced also decrease.

Experts in such domains need prior knowledge to work in such a process and they should be aware of the plant kingdom and its taxonomy. It is because of these problems that automated methods came into existence for identifying and classifying plants. This is a necessity in the field of botany currently.

C. Problem Statement

The amount of farming plants, which in turn depends on the health of these plants, is closely associated with agricultural output. To stop more harm, it is essential to identify and identify the vegetation disease as soon as possible. The identification of illnesses by just sighted the indications on the plant is both simpler and less expensive since macroscopic symptoms may be recognized without the help of a microscope. There are many challenges in general and specific to plant disease detection that needs to be overcome.

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D. Objective of Research

The primary concern addressed in the research on plant disease identification is the development of accurate and efficient methods to detect and diagnose diseases in plants. This is crucial for effective crop management and agricultural practices, as early detection of diseases can assist in averting disease transmission and reducing crop harm, ultimately leading to improved crop yield and food security. Ultimate objective of this research is to build an advanced deep learning algorithm for the identification of plant diseases and providing pesticide recommendations based on a variety of crop images, leveraging the capabilities of EfficientNet.

II. RELATED WORKS

Picon et al. (2019) used a customized deep residual neural network based approach. In order to address the identification of numerous plant diseases under real-time wheat disease detection conditions [4]. With particle swarm optimization, CNN hyperparameters have been improved to classify the model correctly as mentioned by Darwish et al. (2020) [8]. Karthik et al. (2020) used residual learning to learn essential functions and attention mechanisms as input for the identification of disease in tomato leaves. Frequently, deep learning methods are employed for the identification and categorization of plant leaf diseases [9].

Liu et al. (2018) utilized flavia dataset to implement a ten-layer CNN model for the classification of plant leaves and produced the accuracy as 87.92% based on the 32 classes. They used the same model in several training sets based on CNN of remote sensing pictures and achieved a 95% accuracy rate [5]. Bharali et al. (2019), suggested using a tiny CNN to classify plants as healthy or diseased, with a 96.6% accuracy rate. Verma et al. (2020) used the PlantVillage database to categorise the three vine pathologies, comparing the performance of AlexNet with ResNet18. ResNet18 generated a better accuracy rate of 86.43% when compared to AlexNet. Aravind and Raja (2020) employ eggplant, hyacinth bean, ladies finger and lime in six distinct deep learning algorithms: VGG16, DenseNet201 VGG19, ResNet101, GoogleNet AlexNet, and [6].

SVM, textural feature classification, plant leaf image-based classification, K means neural-network-based classification (KMSNN), and segmentation are used in Zhang et al. (2017) proposed approach for classifying cucumber leaf diseases. Carefully extracting and selecting the features from each sick leaf picture is essential, as is calibrating the variables in the SVM and backpropagation network [12].

Pusa Ruby variety of tomato plants was considered for the study of the occurrence of early blight in the plants by Gupta et al (2020) The disease intensity was evaluated with the temperature and relative humidity values. The performance of the stepwise regression model with an RMSE of 6.129 was achieved by them for the prediction of disease intensity [16]. Darwish et al (2020), have used a stepwise regression model to predict late blight in potato plants for the four varieties of potato. The parameters of

temperature, relative humidity, rainfall, and windspeed were considered for disease prediction [8].

Yamamoto et al (2017) suggested a tomato crop disease identification technique in lesions on plant organs. The lesions detection techniques were employed to identify the plant disease in at an early stage. The authors used a super resolution method to improve the quality of the low-resolution images. They compare the performance of different detection techniques with super resolution techniques. Also, they compare the performance of the super resolution methods. The result shows that the super-resolution method accelerates the performance of the image-based phenotyping and vigor diagnosis in the tomato farming than the conventional image scaling methods [11].

Ferentinos (2018) proposed a CNN-based model to detect leaf diseases in 25 distinct plant species. The model was created using the 38 distinct classifications of damaged and healthy plant leaves. The convolutional neural network was trained and evaluated on 87,848 pictures. The created model's average accuracy in test data was 99.53 percent. When the model's performance was compared to other approaches and transfer learning techniques, the authors discovered that it performed significantly better than the other models [2].

Leaf spot infection in sugar beets creates a significant loss in sugar yield. Ozguven & Adem (2019) proposed an image-based automatic leaf spot infection detection in sugar beets. They used faster recurrent CNN architecture to identify the leaf spot disease in sugar beets. The classification accuracy of the model using test images was 95.48 percentage, much higher than conventional approaches and human accuracy. Also, the performance of the faster recurrent CNN architecture was tested in the large agriculture field [7].

A deep learning strategy based on transfer learning was proposed by Zhang et al (2019) to detect *Xanthomonas campestris*-infected peach leaf disease. The peach leaf disease detection algorithm was created with the utilization of refined AlexNet. The algorithm performance was evaluated in comparison to well-known classification methods such SVM, k-NN, and back propagation neural network methods. According to the experiment's findings, AlexNet has been tweaked to reach 100% classification accuracy [13].

Alosaimi et al (2021) presented a convolutional neural network model that identified illnesses in peach plants and fruits. The performance of the trained model was assessed based on the 10-fold cross-validation method. The suggested approach was used to locate the afflicted leaf area and help farmers obtain the right care to preserve the peach crop.

Papaya fruits were categorized by Behera et al (2020), according to their maturity level. A two-stage feature extraction and classification method comprised the machine learning methodology that was employed. The fine-tuning was done utilizing the seven pre-trained CNN networks in this method among which VGG outperformed [14].

Yadav et al (2021) introduced how to identify the disease in Peach plant called bacteriosis using deep learning approaches and imaging scheme, with a dataset having 2705 healthy and defected leaves pictures with an accuracy of 98.75 %. Wani et al., (2022) developed a method for the detection and infections of crops including Rice, Tomato, Potato and Apple. They studied the reasons due to which the crops got diseases and also worked on the possible infections in these crops using aggregate of ML and DL algorithms [10].

III. METHODOLOGY

Fig 2 represents the overall process of plant disease identification utilized in the proposed deep learning model. A general model for detecting plants commences with image capture followed by the enhancing process. Significant features are extracted and then with the help of deep learning algorithms, database matching is performed. It is necessary to consider all the steps in order to obtain maximum accuracy.

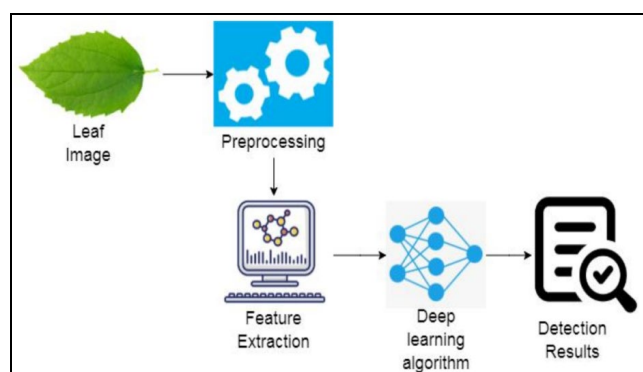


Fig. 2. Plant Detection

The image processing pipeline for disease identification typically involves several key stages those are mentioned Fig.2

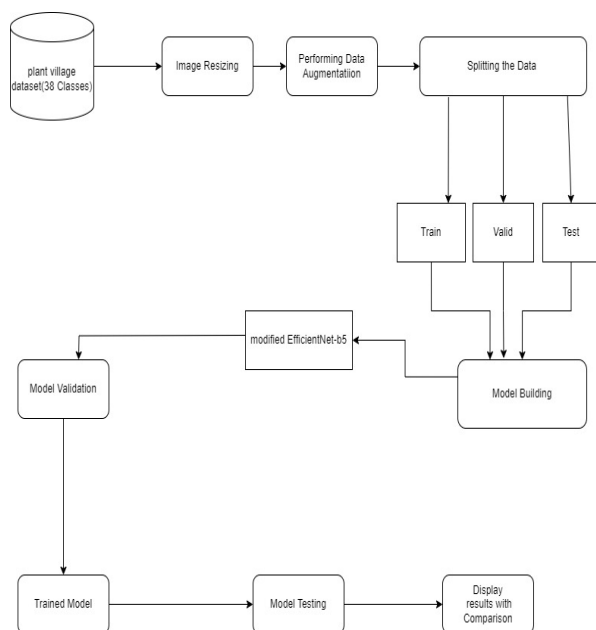


Fig.1.Architecture

A. Data Collection

Fig 1 represents the architectural model of the proposed system. At the initial stage, the image of leaves needs to be collected to make a database. The Plant Village dataset, which comprises 54303 healthy and sick leaf photos categorized into 38 groups based on species and illness, was utilized for this work. Every photograph contained in the Plant Village database was taken at sites for experimental research affiliated with American Land Grant Universities. 54,303 photos representing apple, bell pepper, tomato, potato, squash grape, strawberry, soybean and raspberry are among the 14 crop species were included in the collection.

Images of 12 crop species' healthy leaves were included. There were 38 classifications of good and unhealthy crops in the sample overall. Fig 3 represents the graph which indicates the number of images for each class.

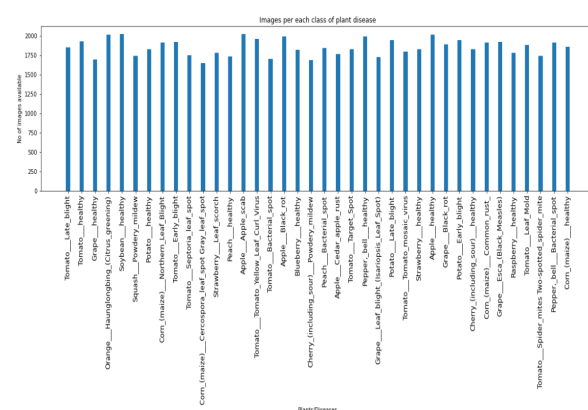


Fig.3. Plant Village Dataset Classes

B. Preprocessing And Data Augmentation

As the plant leaf photos come in different dimensions, we will normalize the images during preprocessing. The pixels in the photos are rescaled during this procedure. The pixels are in the range[0,1] as a result of normalization, which is accomplished by using the rescale parameter in ImageDataGenerator. Besides preprocessing, data augmentation is employed to artificially enhance the diversity of the training dataset. This increased diversity helps the model generalize better to real-world scenarios with varying conditions and ImageDataGenerator class is used for augmentation using the suitable parameters. The parameter controlling zooming is zoom_range. it's set to 0.2, which means that images can be randomly zoomed in by a factor of up to 20%.The parameter controlling horizontal flipping is horizontal_flip. it's set to True, it enables random horizontal flipping of images during training. The parameter controlling rotation is rotation_range it's set to 15, meaning that images can be randomly rotated by up to 15 degrees. The parameter controlling vertical flipping is vertical_flip it's set to True, it enables random vertical flipping of images during training. The parameter controlling brightness adjustment is brightness_range it's set to [0.5, 1.5], allowing for random adjustments to the brightness of images.

C. Features Extraction and Classification

Deep learning enables automated plant disease identification by analyzing image patterns, distinguishing between healthy and diseased plants with high accuracy, aiding in timely intervention for crop protection. The deep learning model EfficientNetB5 is chosen as the base model due to its proven efficiency and effectiveness in image classification tasks. Model size, accuracy, and computing efficiency can be difficult to balance. This problem was addressed with the introduction of the EfficientNet family. To facilitate transfer learning, the model is initially fitted with weights that have already been trained using the ImageNet dataset. This facilitates the extraction of hierarchical features learned from a diverse set of images. All layers of the base model are made trainable to adapt the learned features for the specific task of disease prediction.

The model is fine-tuned on a dataset comprising images of plants affected by various diseases. To decrease the spatial dimensions of the feature maps into a 1D vector, a global average pooling layer is utilized. This step aggregates spatial information and prepares the model for the final classification layer. Batch normalization is applied for normalization, helping stabilize and accelerate the training process. There is also a dropout layer with a 0.2 dropout rate. This lessens the chance of overfitting when training. The final output layer is merged with a set of neurons equivalent to the plant disease classes (38 in this case). The neural network employs the softmax activation layer for the multi-class classification, ensuring output probabilities sum up to 1. During training, the Adam optimizer with a modest learning rate of 0.00001 adjusts effectively. Categorical cross entropy serves as the loss function, effectively measuring the disparity between detected and actual distributions. Evaluation is conducted using accuracy, quantifying the model's ability to classify samples correctly. This setup is robust for training neural networks on multi-class classification tasks. Model predicts the disease and using a defined dictionary (we store the disease and its pesticide as key value pair). It suggests the pesticides and some natural ways to cure diseases.

IV. RESULTS AND DISCUSSION

Fig 4 indicates the accuracy curve and it is steady increasing as the number of training cycle's increases and training and testing accuracy got closely converged at training cycle 10 and it is continued through other cycles this indicates the proposed algorithm ability to simplify well to new data. This was achieved by using Augmentation techniques like scaling and flipping enrich the training dataset, helping the model adapt to diverse plant images and finally got the accuracy 96%.

Table 1 Show in Comparative Analysis. Fig 5 depicts the loss curve, revealing a consistent decline in both training and testing loss as the number of training cycles progresses. Remarkably, the loss curves for both training and testing data converge closely by the 10th training cycle and maintain this convergence throughout subsequent cycles.

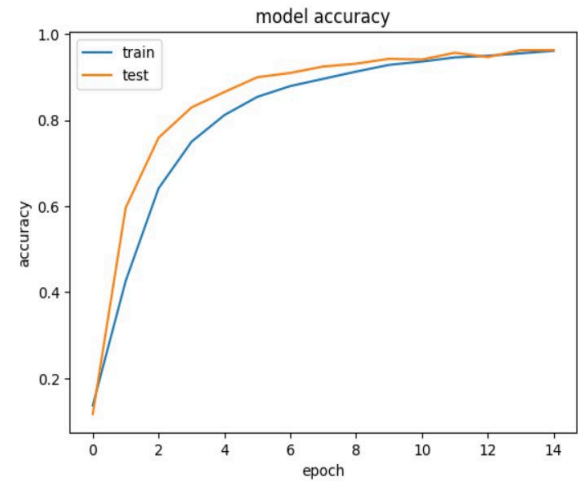


Fig.4. Model Accuracy during Training and Validation

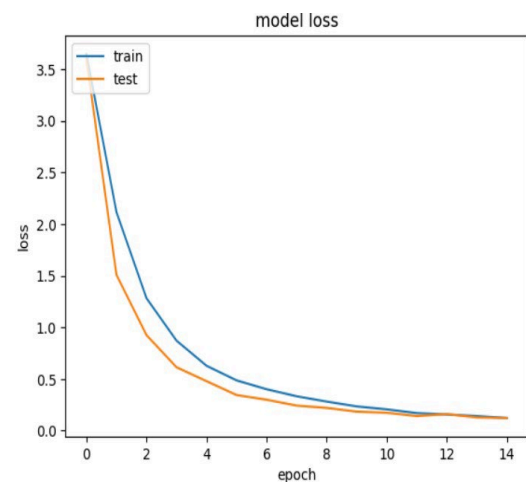


Fig.5. Model Loss during Training and Validation

TABLE 1 COMPARATIVE ANALYSIS

Author	Algorithm	Accuracy	Precision	Recall	F1-Score
Jiang , et al. (2020)	Modified CNN	92	92	93	91
Sachin B. Jadhav, et al. (2020)	DenseNet101	94	94	95	93
Sujatha, et al. (2021)	VGG-16	89.5	89	90	88
Ahmed Elaraby, et al. (2022)	AlexNet	94	94	95	93
Ozguven & Adem (2019)	RCNN	95.48	95	96	94
Proposed EfficientNet	Modified EfficientNet-b5	96	96	96	95

It indicates that errors are getting decreasing for every training cycle and results are getting more accurate. Fig 6 represents the graphical comparative results between proposed modified efficient and previous proposed authors

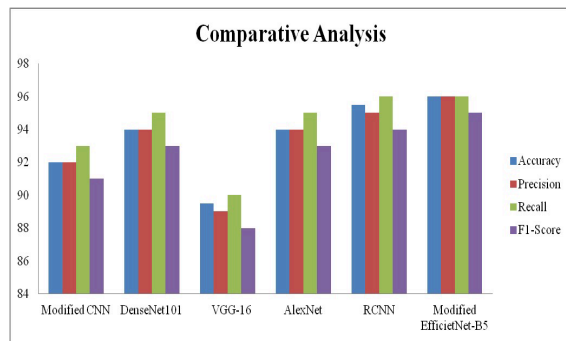


Fig.6. Comparative Analysis

V. CONCLUSION

The early identification of plant diseases is a major challenge to mitigate heavy losses in cultivation. This research focuses on addressing challenges in the identification and diagnosis of plant diseases, particularly those caused by viruses, using deep learning (DL) techniques. By analyzing data on crop production patterns in India across different seasons and regions, the study highlights the significant impact of climate on crop health and productivity. The classification and management of plant diseases not only contribute to agricultural sustainability but also have broader societal benefits by ensuring food security. Comparing the EfficientNet deep learning framework with existing literature, our proposed EfficientNet-B5 model demonstrates promising results, achieving an impressive accuracy of 96% while with reduced training time and offering a compressed model size. This research underscores the potential of DL techniques in revolutionizing plant disease management practices, ultimately benefitting farmers and society as a whole.

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